

6 (**long**) Flow in a pipe of cross-sectional area A_1 undergoes a sudden expansion into a pipe of cross-sectional area A_2 , as shown in Fig. 3. The flow is incompressible with density ρ . Far upstream of the expansion, the flow has uniform velocity V_1 and pressure p_1 . Far downstream of the expansion, the flow has unknown uniform velocity V_2 and pressure p_2 . In the expansion, the flow enters section B as a uniform jet with velocity V_1 , and subsequently undergoes mixing. The pressure in section B can be considered uniform throughout the section. The outside ambient pressure is p_a . Friction with the pipe walls may be neglected throughout.

- (a) Find the velocity downstream of the expansion, V_2 , in terms of V_1 , A_1 and A_2 . [2]
- (b) By considering the control volume that encloses the fluid between sections B and C, or otherwise, find the pressure downstream of the expansion, p_2 , in terms of ρ , p_1 , V_1 , A_1 and A_2 . [12]
- (c) By considering a different control volume, or otherwise, find the magnitude and direction of the force required to restrain the pipe, F . [6]
- (d) Two Pitot tubes connected by a manometer with closed valves containing a column of liquid of density ρ_ℓ are installed upstream and downstream of the contraction, as shown in Fig. 3. If the valves are opened, find the resulting height difference across the manometer, Δh , in terms of ρ , ρ_ℓ , g , p_1 , V_1 , A_1 and A_2 . [10]

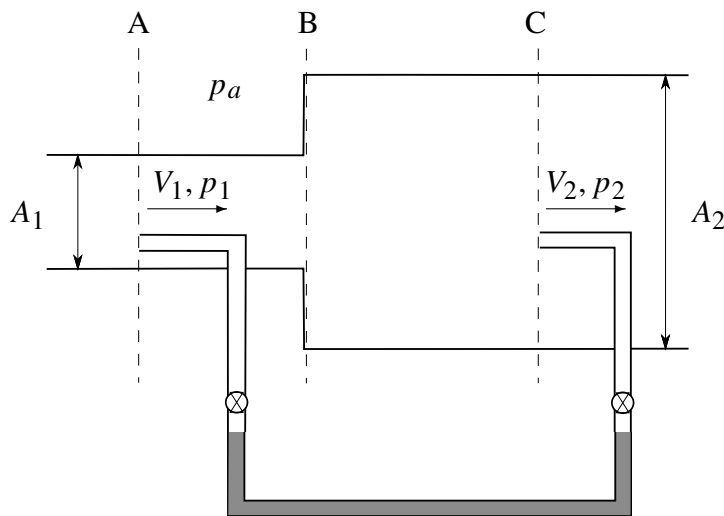


Fig. 3